saas-revenue-analytics

A production-style SaaS revenue analytics warehouse built with dbt Core and DuckDB. Analytics Engineering portfolio project by Arjun Moitra.

Stack	dbt Core 1.12, dbt-duckdb, DuckDB 1.5, Python 3.12, Plotly
Models	6 staging, 8 intermediate, 9 marts (3 dimensions, 4 facts, 2 metric tables)
Tests	344 dbt tests (generic plus 10 singular business-rule tests) and an independent pandas cross-check
Data	Synthetic Stripe-style billing data: 600 customers, 1,080 subscription versions, 5,981 invoices, 6,746 payments, Jan 2023 to Aug 2026

Business problem

Subscription businesses need one trustworthy answer to: what is MRR and ARR right now, how did it move this month, which plan tiers drive revenue and churn, and how much revenue was actually earned versus billed. Billing systems store the raw events but not the answers, and metric definitions drift between spreadsheets. This project puts the definitions in version-controlled, tested SQL and exposes them at grains a BI tool can use directly.

Architecture

```
raw  →  stg_  →  int_  →  dim_ / fct_  →  metrics_  →  dashboard
```

- **Raw:** six CSV extracts loaded untouched into DuckDB (money as cents, dates as text).
- **Staging:** one view per source; rename, cast, trim, normalise statuses and country codes. No business logic.
- **Intermediate:** reusable logic. Subscription versions are classified as new, upgrade, downgrade, seat expansion or contraction, interval change or reactivation using window functions; month-end MRR is built on a date spine; invoice line items are allocated to service months for revenue recognition.
- **Marts:** dim_customers, dim_plans, dim_subscriptions, fct_invoices, fct_payments, fct_revenue (customer x plan x month), fct_mrr_movements (customer x month), metrics_monthly and metrics_monthly_by_plan.
- **Dashboard:** a single-file Plotly HTML page that reads only from the marts: executive tiles, MRR waterfall, recognized vs billed revenue, churn and retention, ARPU, plan-tier mix, cohort retention and a table view.

Key metric definitions

Metric	Definition
MRR	Sum of unit price x seats for paid subscription versions active on the last day of the month; annual prices divided by 12. Trials, coupons and one-time fees excluded.
ARR	MRR x 12. A run-rate, not a forecast.
New / expansion / contraction / churned / reactivation MRR	Month-over-month movement per customer. Enforced identity: mrr = beginning + new + expansion + reactivation - contraction - churned.
Customer churn rate	Churned customers / prior month active customers.
ARPU	MRR / active customers.
Recognized revenue	Each invoice line spread evenly by day across its service period (simplified, clearly labelled, not ASC 606 advice).
Deferred revenue	Cumulative billed (ex-tax) minus cumulative recognized.

Data quality strategy

- Primary keys tested unique and not null; composite grains enforced with unique_combination_of_columns.

- Relationships from every fact foreign key to its dimension; accepted_values on every status column; accepted ranges on rates and seats; a custom non_negative test on every money and count column.
- Singular tests for the rules a finance reviewer checks by hand: invoice totals reconcile to line items, payments never exceed invoice totals, subscription versions never overlap, the MRR waterfall identity holds for every customer-month, recognized revenue sums back to each line to the cent, trials never bill.
- scripts/validate_metrics.py recomputes MRR, ARR, ARPU and active customers from the raw CSVs in pandas and compares them to the marts.

Example insights from the snapshot month (Aug 2026)

- ARR is $8.40M, up 57% year over year, driven by new logos (295 to 454 active customers) while ARPU held near $1.5K.
- Enterprise is 10% of customers and 57% of MRR; the top 10 accounts are 23% of ARR.
- Churn is concentrated in the low tiers: 1.8% monthly on Starter versus 0% on Enterprise.
- Net revenue retention averages 99% monthly: expansion roughly offsets contraction and churn but does not exceed it.
- Billed revenue swings between $426K and $898K because 52% of billings are annual, while recognized revenue climbs steadily to $705K; deferred revenue stands at $2.2M.

Run it locally

```
make setup      # venv + dbt-core, dbt-duckdb
make data       # generate CSVs, load raw schema
make build      # dbt clean + deps + build (models and tests)
make validate   # independent cross-check
make dashboard  # writes dashboard/index.html
make docs       # dbt docs generate + serve
```

Simplifications, kept deliberately: synthetic single-currency data, plan changes only on billing period boundaries (no proration), list-price MRR, straight-line revenue recognition, monthly rather than trailing-12 retention, full-refresh marts. See README.md for full documentation.